

Bioinformatics_Albinism.docx

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Oculocutaneous Albinism Type IA

1 Introduction

Albinism is an autosomal recessive genetic disorder characterised by the lack or reduction of melanin production in melanocytes (Marçon & Maia, 2019). It is caused by the genetic defect of an enzyme called tyrosinase, where the enzyme is responsible for melanin pigment production. There are two main types of albinism: oculocutaneous albinism (OCA) and ocular albinism (OA). In OCA, the absence or incomplete melanin biosynthesis causes ocular abnormalities and hypopigmentation in the eyes, skin, and hair. In contrast, in OA, the visual pathway is the only one clinically impacted. There are eight forms of OCA have been recognized: OCA1, OCA2, OCA3, OCA4, OCA5, OCA6, OCA7, and OCA8, in which OCA1 is divided into subtypes OCA1A and OCA1B.

In this bioinformatics investigation of albinism, I focused on the most severe type, which is OCA1A, tyrosinase-negative oculocutaneous albinism. In OCA1A, the activity of tyrosinase is entirely inactive. This results from a tyrosinase gene (*Tyr*; [606933.0001](#)) mutation that is either homozygous or compound heterozygous. Patients have a permanent lack of melanin pigment from birth and, therefore, have a heightened vulnerability to UV radiation and a higher risk of developing skin cancer. In addition, visual abnormalities such as reduced visual acuity, strabismus, nystagmus, and photophobia are also the consequences brought by OCA1A, as stated in the clinical synopsis section on the Online Mendelian Inheritance in Man ([OMIM](#)) database.

this is a good introduction. Try to cite all your sources.

2 *Tyr* gene

The *tyr* gene codes the enzyme tyrosinase. The information of it can be found on the National Centre for Biotechnology Information ([NCBI](#)) website, showing that *Tyr* is located on the chromosome 11: q14.3 in *Homo sapiens* (Figure 2.1). The *Tyr* gene has a MIM number of 606933, which means we can also obtain information on *Tyr* genes, such as gene function and animal models, on [OMIM](#) by using this unique number.

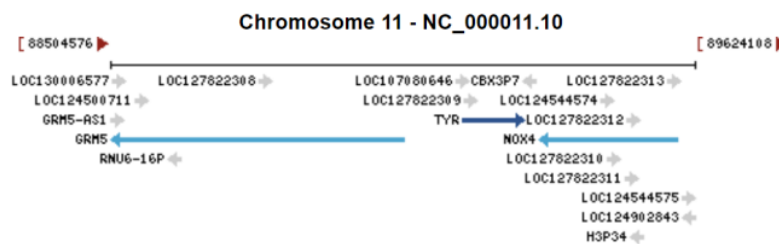


Fig. 2.1 An overall view of chromosome 11 and the relative position of the *tyr* gene. (Sayers et al., 2021)

On the [genome data viewer](#) of NCBI, I chose the primary assembly GRCh38.p14, and its graphical alignment is shown in Figure 2.2. From the genome data viewer, I found five exons and four introns in the tyrosinase gene, and the exon-intron boundaries match those of the mouse gene. The second purple segment has the reference sequence [NM_000372.5](#). It is the mRNA of the protein tyrosinase (Fig. 2.3).

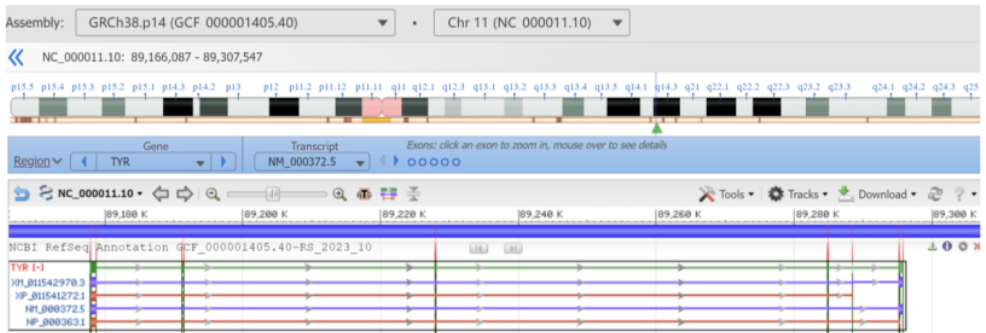


Fig. 2.2 Graphical view of assembly GRCh38.p14.

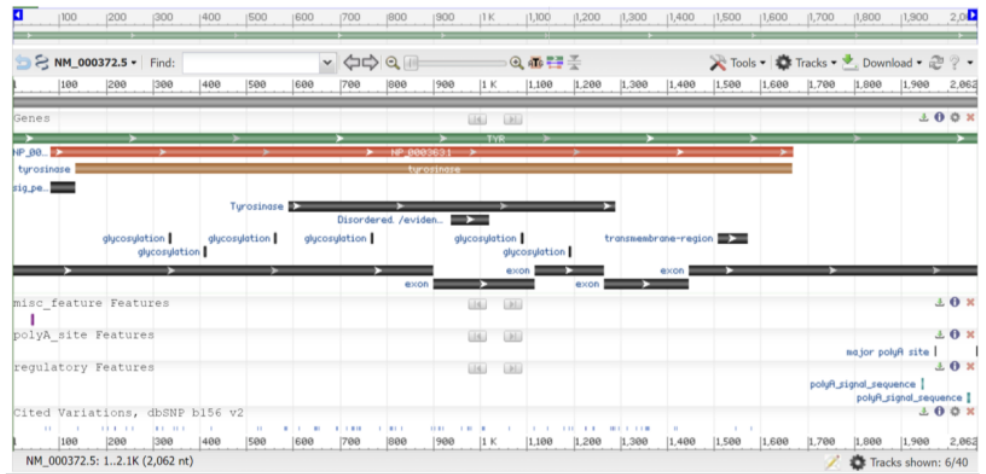


Fig. 2.3 Graphical view of Homo sapiens tyrosinase (TYR), mRNA, where cited variations can be found.

what are these variations? Highlight them and elaborate in the legend.

3 Tyrosinase

Tyrosinase is an oxidase that contains copper that is precisely located on the melanosome membrane, so it involves reactions both across the membrane and inside the melanosome lumen.^{[citation?](#)} It is responsible for synthesising pigments like melanin and other polyphenolic substances. Different pathway information of tyrosinase can be found on [Pathway Commons](#) (Anon, n.d.). The website [UniProt](#) is a database of proteins (The UniProt Consortium, 2023). From this, I found the three-dimensional structure of the tyrosinase protein from [AlphaFold](#), which is shown in Fig.3.1. It contains a copper-binding domain, which can bind two copper ions. This copper-binding domain plays a crucial role in melanin biosynthesis.

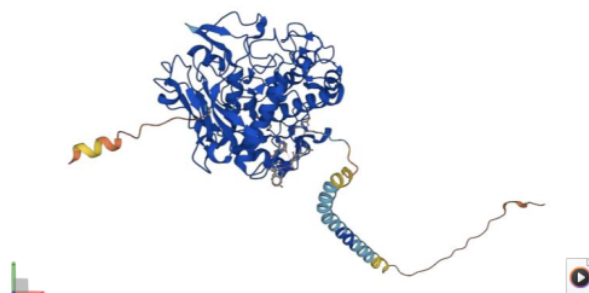


Fig. 3.1 A dimension of the 3D structure of tyrosinase enzyme. (Jumper et al., 2021; Varadi et al., 2022)

Information on melanin biosynthesis can be found on the website [Reactome](#), and Fig 3.2 below summarises the entire process of synthesising melanin. Here, tyrosinase catalyses the first and rate-limiting step in the series of events that produce melanin from tyrosine. It first converts tyrosine into DOPA (3,4-dihydroxyphenylalanine) by hydroxylation, then facilitates the conversion of DOPA into DOPA-quinone ($2 \text{ L-dopa} + \text{O}_2 = 2 \text{ H}_2\text{O} + 2 \text{ L-dopaquinone}$), the overall chemical equation that tyrosinase catalyses is $\text{L-tyrosine} + \text{O}_2 = \text{H}_2\text{O} + \text{L-dopaquinone}$.

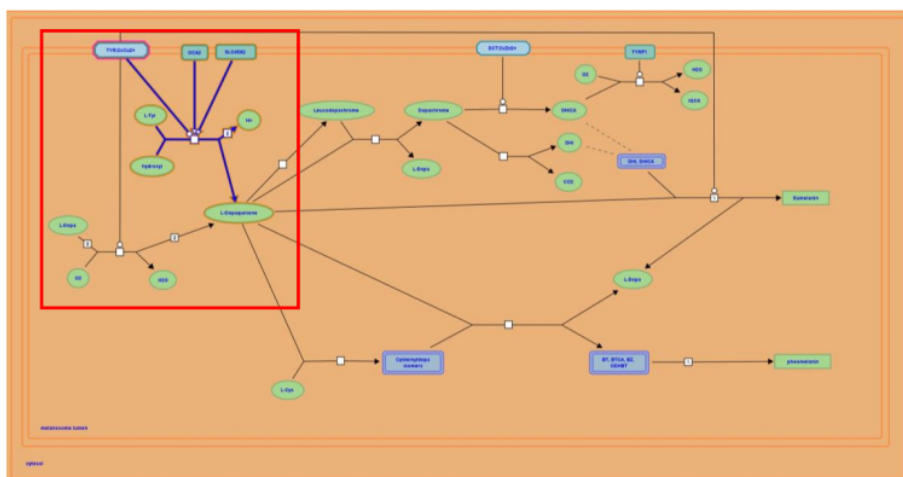


Figure 3.2 Steps of melanin biosynthesis: tyrosinase oxidises tyrosine to dopaquinone; dopaquinone transforms

I would blow up this figure to make it legible.

to leucodopachrome (cyclodopa); dopa is oxidised to dopaquinone by tyrosinase; dopaquinone, cysteine form CD isomers; CD isomers transform to BT, BTCA, BZ, ODHBT; pheomelanin formation; dopachrome is transformed to DHICA by DCT; TYRP1 oxidises DHICA to IQCA; dopachrome transforms to DHI; dopachrome transforms to DHI; DHI and DHICA polymerise forming eumelanin. (Jupe, 2015)

As tyrosinase is a copper-dependent enzyme, it requires the co-expression of other genes, such as the ATP7A gene, to work. ATP7A facilitates the transportation of copper into the secretory pathway of mammalian cells (Petris, Strausak & Mercer, 2000). In addition, ATP7A is involved in reloading copper to melanosomes for tyrosinase to use (Anon, n.d.).

Inactive tyrosinase is found in patients with OCA1A. Research has proposed that a homozygous 1-bp insertion (C) is located between nucleotides 1011 and 1012 in a kid with OCA1A. A premature termination signal (TGA codon) was added after amino acid residue 298 by the insertion, which also changed the reading frame. As a result, the enzyme was shortened and lost one of its two copper-binding domains. According to functional studies, the shortened tyrosinase lacked catalytic activity. Since their DNA reacted with both the mutant probe and the normal probe, the parents and one sibling were heterozygous (Chintamaneni et al., 1991). This variant will be discussed in the next section. *this was a well-written section, well done. Always support your arguments with additional citations?*

4 Variants

There are, in total, [513](#) related variants for the *Tyr* gene on NCBI, 104 of which will lead to OCA1A (Landrum et al., 2018). According to the [Ensembl](#) website, one of the variants for OCA1A has a reference number of rs281865527, and it is found on the exon 2 of the *Tyr* gene, as shown in Fig. 4.1.

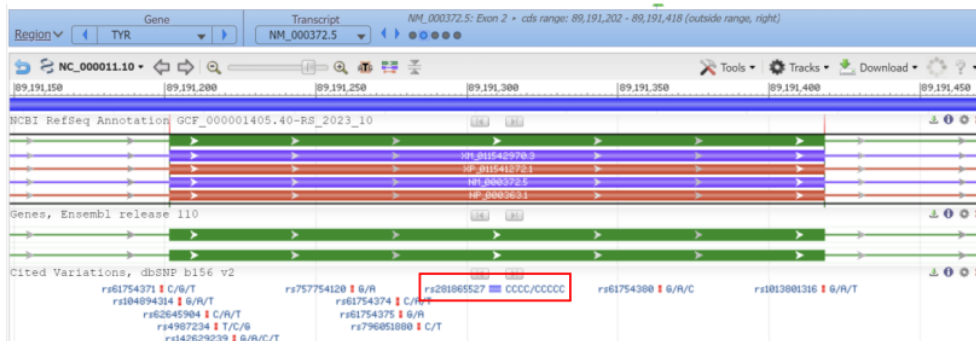


Fig. 4.1 Exon 2 is shown, and the variant rs28186552 is highlighted in the red box.

The variant is found to be pathogenic on [ClinVar](#). It is a frameshift variant because the number of nucleotides added or removed is not a multiple of three, which disrupts the translational reading frame. Indels of the *Tyr* gene create this variant [rs281865527](#). Extra duplication of cytosine is found. CCCC/CCCCC is the allele for this variant. This means that on the codon, there should be ACC CCA, for example, but now it has ACC CCC A instead (Fig. 4.2). The [dbSNP](#) database on NCBI shows the same information that there is an insertion of cytosine in the amino acid codon. The

very well done. This sections shows some good analytical skills. Use this section as an example on how to analyse information from various sources.

flanking sequence of this specific allele also can be found on the [Ensembl](#) webpage (Martin et al., 2023).

The insertion added a premature termination signal (TGA codon) after amino acid residue 298, changing the reading frame. As a result, the enzyme was reduced in length, one of its two copper-binding domains was destroyed, and the truncated tyrosinase lacked catalytic activity.

Genes and regulation ⓘ

Gene and Transcript consequences

Show/hide columns

Filter

Gene	Transcript (strand)	Allele (Tr. allele)	Consequence Type	Position in transcript	Position in CDS	Position in protein	AA	Codons	Detail
ENSG00000077498 HGNC: TYR	ENST00000526139.1 (+) biotype: protein_coding_CDS_not_defined	CCCCC (CCCCC)	non coding transcript exon variant	987-990 (out of 1446) 	-	-	-	-	Show
ENSG00000077498 HGNC: TYR	ENST00000263321.6 (+) biotype: protein_coding	CCCCC (CCCCC)	frameshift variant	1005-1008 (out of 2062) 	926-929 (out of 1590) 	309-310 (out of 529) 	TP/TPX	ACCCCA/ACCCCA	Show

Fig. 4.2 Information for the variant rs281865527 from Ensembl.

[NCBI ALFA](#) has collected data about population genetics. It shows that East Asians have the highest CCCCC allele compared to *Homo sapiens* from other regions (Fig 4.3). The same data is displayed on [Ensembl](#).

Population	Group	Sample Size	Ref Allele	Alt Allele
Total	Global	85874	CCCC=0.99997	CCCCC=0.00003
European	Sub	75536	CCCC=0.99999	CCCCC=0.00001
African	Sub	3698	CCCC=1.0000	CCCCC=0.0000
African Others	Sub	166	CCCC=1.000	CCCCC=0.000
African American	Sub	3532	CCCC=1.0000	CCCCC=0.0000
Asian	Sub	3272	CCCC=0.9994	CCCCC=0.0006
East Asian	Sub	2646	CCCC=0.9992	CCCCC=0.0008
Other Asian	Sub	626	CCCC=1.000	CCCCC=0.000
Latin American 1	Sub	436	CCCC=1.000	CCCCC=0.000
Latin American 2	Sub	928	CCCC=1.000	CCCCC=0.000
South Asian	Sub	274	CCCC=1.000	CCCCC=0.000
Other	Sub	1730	CCCC=1.0000	CCCCC=0.0000

Fig. 4.3 Population genetics data of the mutated CCCCC allele from ALFA.

this is a table. It also needs more explanation in the legend.

Apart from that, on [Uniprot](#), there are many other variants for OCA1A. For example, the Var_007651 variant has the reference number [rs28940878](#). Unlike the variant I found on OMIM that causes frameshift by duplication of cytosines, this variant causes missense mutation, which means that an alteration of nucleotides causes a change in the amino acid sequence. Adenosine is mutated into guanine, so one of the codons that is involved in tyrosinase synthesis is substituted from GAG into GGC, as shown in Fig.4.5. In addition, Fig. shows the graphical diagram where this variant is located, which is at the position [11:89178028-89178128](#).*which figure? A good proofreading before submission would help you avoid such typos*

Genes and regulation

Gene and Transcript consequences

Show/hide columns										Filter	
Gene	Transcript (strand)	Allele (Tr. allele)	Consequence Type	Position in transcript	Position in CDS	Position in protein	A A	Codons	SIFT	PolyPhen	
ENSG00000077498 HGNC: TYR	ENST00000526139.1 (+) biotype: protein_coding_CDS_not_defined	G (G)	non coding transcript exon variant	186 (out of 1446)	-	-	-	-	-	-	
ENSG00000077498 HGNC: TYR	ENST00000263321.6 (+) biotype: protein_coding	G (G)	missense variant	204 (out of 2062)	125 (out of 1590)	42 (out of 529)	D/G	GAC/GGC	0.02	0.999	

Fig. 4.5 Information for the variant rs28940878 from Ensembl.

this would be a new section with the heading population genetics. It would need, apart from the table below, some analysis.

To investigate the population genetics of variant Var_007651, we can use the [NCBI ALFA database](#). It shows that African Americans and Africans show the greatest count of allele frequency.

Population	Group	Sample Size	Ref Allele	Alt Allele
Total	Global	654	A=0.998	G=0.002
European	Sub	76	A=1.00	G=0.00
African	Sub	434	A=0.998	G=0.002
African Others	Sub	0	A=0	G=0
African American	Sub	434	A=0.998	G=0.002
Asian	Sub	30	A=1.00	G=0.00
East Asian	Sub	30	A=1.00	G=0.00
Other Asian	Sub	0	A=0	G=0
Latin American 1	Sub	0	A=0	G=0
Latin American 2	Sub	0	A=0	G=0
South Asian	Sub	6	A=1.0	G=0.0

Fig. 4.6 Population genetics data of variant rs28940878 from ALFA database.

5 Animal model

OCA1A is not shown only in *Homo sapiens* but also in other numerous numbers of animals. On the website [OMIA](#), it shows the phenotypes, species names, and their variants. Take [Oryctolagus cuniculus](#) as an example; the coat colour oculocutaneous albinism type I is caused by a missense mutation from tyrosine into lysine at codon 373 that is similar to OCA1A in humans.

Moreover, the information on homologous genes is found on HomoloGene. For example, *Saccharomyces cerevisiae* or Brewer's yeast has a [homologous gene 40118](#). I did a pairwise alignment between *Saccharomyces cerevisiae* and another yeast called *Kluyveromyces lactis* using [Blast](#) and got an E-value of $2e^{-124}$ and a percentage identity of 82.47%. The alignment is shown in Fig. 5.1. *why would S cerevisiae have an ablinism-related protein? This is a homologous protein and the organisms are too distant between them. A pairwise comparison between two mammals would be more beneficial for this essay.*

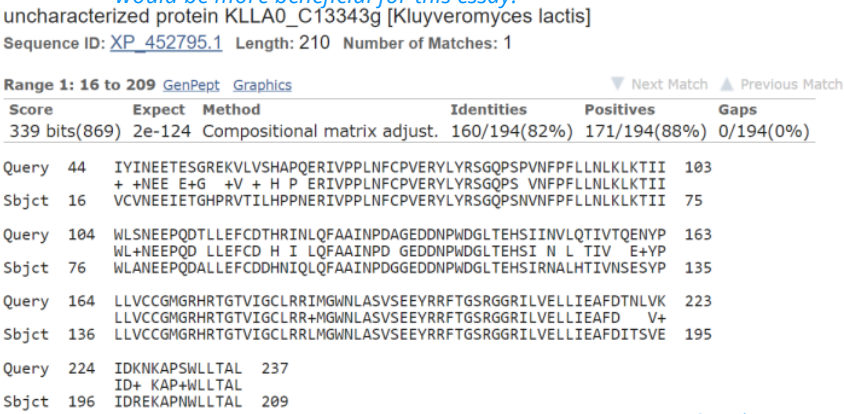


Fig. 5.1 An alignment *Kluyveromyces lactis* gene, blasted with a homologous gene in *Saccharomyces cerevisiae*. *this alignment does not provide any information on animal models.*

6 Conclusion

In conclusion, OCA1A has been explored using various bioinformatics databases and tools, focusing on genes, proteins, and variants. Intricate molecular underpinnings of this genetic disorder have been revealed. By leveraging the resources provided by NCBI and OMIM databases, the genetic variations and protein alterations that contribute to the pathogenesis of albinism have been discovered. Specifically, looking into two variants of *tyr* genes explained how the mutation of *tyr* leads to disrupted melanin synthesis and OCA1A.

This in-depth analysis not only deepens the comprehension of the genetic basis of OCA1A but also paves the way for potential therapeutic interventions that target the underlying molecular mechanisms. More effective treatments will be discovered, accompanied by the continued research efforts and the utilisation of these comprehensive bioinformatics tools.

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FINAL GRADE

GENERAL COMMENTS

57 /100

PAGE 1

Text Comment. citations?

Text Comment. this is a good introduction. Try to cite all your sources.

PAGE 2

Text Comment. what are these variations? Highlight them and elaborate in the legend.

PAGE 3

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PAGE 4

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PAGE 5

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PAGE 6

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PAGE 7

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PAGE 8

PAGE 9

GRADING FORM: LIFESCIGRADINGFORMMINORV1

YUKI WU

CRITERIA MET

How does this work meet the criteria for the grade it has been given?



The student has used bioinformatic tools and websites beyond the ones taught during the practicals.

Student has used information from different sources to build arguments using analytical thinking in a couple of occasions.

AREAS TO IMPROVE

How could this work – and similar work in future – be improved, to gain a higher mark?



You might want to focus on the structure of your essay and the links between paragraphs to obtain a better flow.

Also proofread before submitting to correct mistakes and typos.

Legends of figures need to be more explanatory.

In order to obtain higher marks in essays in the future, try to critically synthesise the information you obtain from papers, rather than being descriptive.